MATHEMATICS CLASS – X WBBSE 2026

1. **Choose the correct option in each case from the following questions:** (1X6=6)

i. **If a principal becomes double in 10yrs, the rate of interest per annum is:**
(a) 5% (b) 10% (c) 15% (d) 20%

ii. **Condition of two roots of a quadratic equation $ax^2 + bx + c = 0, a > 0$ will be equal in magnitude but opposite in sign:**
(a) b = 0, c = 0 (b) b = 0, c > 0 (c) b = 0, c < 0 (d) b > 0, c = 0

iii. **If average 6, 7, x, y, 16 be 9, then**
(a) $x + y = 21$ (b) $x + y = 16$ (c) $x - y = 21$ (d) $x - y = 19$

iv. **Arc of length 121 cm of a circle makes 77^0 angle at the centre of the circle then the radius of the circle will be:**
(a) 110 cm (b) 100 cm (c) 90 cm (d) 70 cm

v. **Length of a side of a cube be a unit and length of the diagonal be d unit then the relation of a and d will be:**
(a) $\sqrt{2}a = d$ (b) $\sqrt{3}a = d$ (c) $a = \sqrt{3}d$ (d) $a = \sqrt{2}d$

vi. **ABCD be a cyclic quadrilateral whose centre is O, BC is extended upto E. If $\angle DCE = 96^0$, then the value of $\angle BOD$ will be:**
(a) 42^0 (b) 84^0 (c) 142^0 (d) 168^0

2. **Fill in the blanks (any Five):-** (1X5=5)

i. **If the ratio of principal and yearly amount be 8:9, then yearly rate of interest is ____________ .**

ii. **Conjugate surd of $(\sqrt{3} - 5)$ is ____________ .**

iii. **Two tangents at the end of a diameter of a circle are mutually ____________**

iv. **If $x = a \sec\theta$ and $y = b \cot\theta$ then $\frac{x^2}{a^2} - \frac{b^2}{y^2} =$ ____________**

v. **The radius of a solid hemisphere is 3r unit, the area of total surface is ____________ .**

vi. **The frequency of 1, 2, 3, 4, 5 are respectively 1, 2, 3, 4, f and their arithmetic mean is 4 then the value of f is ____________**

3. **Write True or False (any Five):-** (1X5=5)

i. $\sin^2\theta = (\sin\theta)^2, 0^0 < \theta < 90^0$.

ii. **The length of a largest cube be $4\sqrt{2}$ cm inscribed in a sphere of radius 4 cm.**

iii. **The angle in the segment of a circle which is greater than semicircle is an obtuse angle.**

iv. **If the Arithmetic mean of x-3, x-1, x, 2x-1. 3x-5 be 7.5, then their median will be 3.**

v. **If $x \propto \frac{1}{y}$, then $(xy)^{10}$ is a constant.**

vi. **In a business, the ratio of capital of Raju and Asif is 5:4. If Raju got Rs 80 of total profit then Asif got Rs. 100.**

4. Answer the following questions (any ten):- (2X10=20)

i. **A and B started a business by investing Rs. 15,000 and Rs. 45,000 respectively. After 6 months B got a profit of Rs. 3,030, what is the profit of A?**

ii. **In a triangle ABC, a straight line parallel to BC intersects AB and AC respectively at P and Q. If AP = 4 cm, QC = 9 cm and if PB = AQ then find the value of PB.**

iii. **Two chords AB and CD are equidistant from the centre of a circle O. If $\angle AOB = 60^0$ and CD = 6 cm, find the radius of the circle.**

iv. **If $\tan\theta + \cot\theta = 2$, then the value of $\tan^7\theta + \cot^7\theta$.**

v. **If x and y are positive real numbers then $\sec\theta = \frac{x}{y}$ is correct or not? Give answer with reason.**

vi. **For two right circular cylinders, if ratio of their heights be 1:2 and the ratio of the circumference of the base be 3:4, then find the ratio of their volume.**

vii. **Arithmetic mean of $x_1, x_2, x_3, \dots, x_n$ is $\bar{x}$. Prove that $\sum_{i=1}^{n}(x_i - \bar{x})^2 = \sum_{i=1}^{n}(x_i)^2 - n(\bar{x})^2$.**

viii. **If the rate of interest increases from 5.5% to 6% then the yearly interest increased by Rs. 49.50. Find the capital.**

ix. **If the sum of the roots of the equation $x^2 - 4x = K(x-1) - 5$ is 7, then find the value of K.**

x. **If $(a+b):\sqrt{ab} = 2:1$, then find a:b.**

xi. **If the radius of a sphere be increased by 50%, find the percentage increased of the volume.**

xii. **ABCD be a cyclic quadrilateral. If AD = AB, $\angle DAC = 60^0$ and $\angle BDC = 50^0$ then find the $\angle ACD$.**

5. Answer any one question:- (1X5=5)

i. **If the rate of compound interest be 4% in the 1st year and 5% in the 2nd year, then find the interest of Rs. 25,000 for two years.**

ii. **Three friends invest Rs. 4,800, Rs. 6,600 and Rs. 9,600 respectively in a business. 1st person received $\frac{1}{8}$ th of the profit as salary for looking after the business and the remaining profit was distributed among them in the ratio of their capitals. If after one year 1st person received Rs. 780 find the amount received by other two.**

6. Solve any one question:- (1X3=3)

i. **Solve $b(c-a)x^2 + c(a-b)x + a(b-c) = 0$.**

ii. **Digit in the ten's place of a two digit number is less by 3 than the digit in the unit place. Product of the digits is less than the number by 15. Find the number.**

7. Answer any one question:- **(1X3=3)**

i. If $(x^3 + y^3) \propto (x^3 - y^3)$, then prove that $(x^2 + y^2) \propto xy$

ii. If $x(2-\sqrt{3}) = y(2+\sqrt{3}) = 1$, then the value of $3x^2 - 5xy + 3y^2$.

8. Answer any one question:- **(1X3=3)**

i. If $\frac{a+b-c}{a+b} = \frac{b+c-a}{b+c} = \frac{c+a-b}{c+a}$ and $a + b + c \neq 0$, then prove that $a = b = c$.

ii. If $x = \frac{8ab}{a+b}$, then find the value of $\frac{x+4a}{x-4a} + \frac{x+4b}{x-4b}$.

9. Answer any one question:- **(1X5=5)**

i. Prove that the front angle formed at the centre of the circle by an arc is double of the angle formed by the same arc at any point of the circle.

ii. If two circles touch each other, prove that the point of contact lies on the line joining the centre of two circles.

10. Answer any one question:- **(1X3=3)**

i. In the isosceles triangle ABC, $\angle B$ is a right angle. Bisector of the $\angle BAC$ intersects BC at D. Prove that $CD^2 = 2BD^2$.

ii. O is a point inside a rectangle ABCD. Prove that $OA^2 + OC^2 = OB^2 + OB^2$.

11. Answer any one question:- **(1X5=5)**

i. In triangle ABC the base BC = 6 cm, $\angle ABC = 60^0$ and AB = 8 cm. Draw the circumcircle of the triangle.

ii. Construct a square of equal area of an equilateral triangle of side 6 cm..

12. Answer any two questions:- **(3X2=6)**

i. If the ratio of angles of a triagle is 2:3:4 then determine the circular value of the greatest side.

ii. If $tan\ \theta = \frac{4}{3}$, find the value of $sinn\ \theta + cos\ \theta$.

iii. If A and B are complementary angles then prove that:

$$(sin\ A + cos\ B)^2 = 1 + 2sin\ A\ sin\ B$$

13. Answer any one question:- **(1X5=5)**

i. From the roof of a building the angle of depression of the top and foot of a lamp post are 30^0 and θ^0 respectively. If the ratio of the height of the building and the height of the lamp post be 3:2, then find the value of θ.

ii. From the foot of a Tilla the angle of elevation of its top is 45^0. By moving 100 m towards the Tilla along a slope of 30^0, the angle of elevation of the top becomes 60^0. Find the height of the Tilla.

14. Answer any two questions:- **(4X2=8)**

i. The ratio of the length, breadth and height of a solid rectangular parallelepiped is 4:3:2 and area of the whole surface is 468 sq cm. Find the volume of the parallelepiped.

ii. **The internal and external radius of a hollow cylinder of height 20 cm are 4 cm and 5 cm respectively. By melting a solid cone of height equal to one third height of the cylinder is formed. Find the diameter of the base of the cone.**

iii. **A hemisphere bowl of internal radius 9 cm is full of water. How many cylindrical bottle of diameter 3 cm and height 4 cm are required to fill the water?**

15. Answer any two questions:- **(4X2=8)**

i. **Find the arithmetic mean of the following distribution:**

Class	5-14	15-24	25-34	35-44	45-54	55-64
Frequency	3	6	18	20	10	3

ii. **Making cumulative frequency (greater than type) distribution table of given data, draw Ogive on graph paper.**

Class	100-120	120-140	140-160	160-180	180-200
Frequency	8	14	10	12	4

iii. **Find the mode of the following frequency distribution:**

Marks Obtained	Less than 10	Less than 20	Less than 30	Less than 40	Less than 50	Less than 60
Number of Students	8	13	29	42	60	70

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